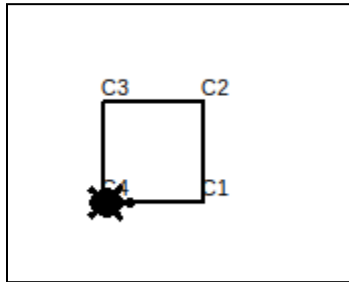
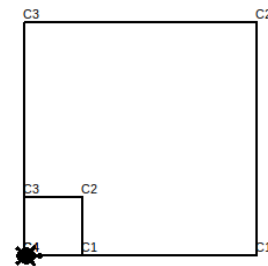


Prelab Part 1

- action/client file can be found in ~/Lab4/chuang/intro_ws/src/turtle_test/



```
yahboom@yahboom-virtual-machine: ~/Desktop/S26_RobotP...  
[INFO] [1773449381.740964475] [square_client]: Goal succeeded! Final pose: turtle_interfaces.msg.TurtlePose(heading=0.0, position=array([-9.1848509e-15, 7.1054274e-15], dtype=float32))  
yahboom@yahboom-virtual-machine: ~/Desktop/S26_RobotProgramming_Iron/Lab4/chuang/intro_ws$ ros2 run turtle_test make_square --ros-args --param square_size:=200.0  
[INFO] [1773449467.765764915] [square_client]: Waiting for action server...  
[INFO] [1773449467.766498778] [square_client]: Sending goal request...  
[INFO] [1773449467.768918428] [square_client]: Goal accepted  
[INFO] [1773449468.860653599] [square_client]: Feedback: turtle_interfaces.msg.TurtlePose(heading=90.0, position=array([ 2.0000000e+02, -4.1880443e-14], dtype=float32))  
[INFO] [1773449470.956138754] [square_client]: Feedback: turtle_interfaces.msg.TurtlePose(heading=180.0, position=array([200., 200.], dtype=float32))  
[INFO] [1773449473.044703308] [square_client]: Feedback: turtle_interfaces.msg.TurtlePose(heading=270.0, position=array([5.684342e-14, 2.000000e+02], dtype=float32))  
[INFO] [1773449475.165659988] [square_client]: Feedback: turtle_interfaces.msg.TurtlePose(heading=0.0, position=array([-2.888186e-14, 5.684342e-14], dtype=float32))  
[INFO] [1773449476.168502102] [square_client]: Goal succeeded! Final pose: turtle_interfaces.msg.TurtlePose(heading=0.0, position=array([-2.888186e-14, 5.684342e-14], dtype=float32))  
yahboom@yahboom-virtual-machine: ~/Desktop/S26_RobotProgramming_Iron/Lab4/chuang/intro_ws$
```



Prelab Part 2

- Modified launch file can be found in ~/Lab4/chuang/intro_ws/src/python_parameters

unmodified

```

minimal_param_node-1] [INFO] [1773452552.379869817] [group6.Hiro]: Hello Hiro!
minimal_param_node-1] [INFO] [1773452553.369773195] [group6.Hiro]: Hello Hiro!
minimal_param_node-3] [INFO] [1773452553.875123069] [group0.NA]: Hello NA!
minimal_param_node-1] [INFO] [1773452554.369767422] [group6.Hiro]: Hello Hiro!
minimal_param_node-1] [INFO] [1773452555.369428814] [group6.Hiro]: Hello Hiro!
minimal_param_node-3] [INFO] [1773452556.364533038] [group0.NA]: Hello NA!
minimal_param_node-1] [INFO] [1773452556.369497441] [group6.Hiro]: Hello Hiro!
minimal_param_node-1] [INFO] [1773452557.369708740] [group6.Hiro]: Hello Hiro!
minimal_param_node-1] [INFO] [1773452558.369695565] [group6.Hiro]: Hello Hiro!
minimal_param_node-3] [INFO] [1773452558.865213181] [group0.NA]: Hello NA!
minimal_param_node-1] [INFO] [1773452559.369696605] [group6.Hiro]: Hello Hiro!

```

modified

```

yahboom@yahboom-virtual-machine:~/Desktop/S26_RobotProgramming_Iron/Lab4/chuang/
intro_ws$ ros2 launch python_parameters run_multi_launch.py my_parameter:=Ian
[INFO] [launch]: All log files can be found below /home/yahboom/.ros/log/2026-03
-14-09-40-34-643207-yahboom-virtual-machine-23846
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [minimal_param_node-1]: process started with pid [23847]
[INFO] [minimal_param_node-2]: process started with pid [23849]
[INFO] [minimal_param_node-3]: process started with pid [23851]
[INFO] [minimal_param_node-3]: process started with pid [23851]
minimal_param_node-1] [INFO] [1773452435.998912185] [group6.Hiro]: Hello Hiro!
minimal_param_node-1] [INFO] [1773452436.988238071] [group6.Hiro]: Hello Hiro!
minimal_param_node-3] [INFO] [1773452437.490646447] [group0.NA]: Hello Ian!
minimal_param_node-1] [INFO] [1773452437.988335283] [group6.Hiro]: Hello Hiro!
minimal_param_node-1] [INFO] [1773452438.988776373] [group6.Hiro]: Hello Hiro!
minimal_param_node-3] [INFO] [1773452439.979002499] [group0.NA]: Hello Ian!
minimal_param_node-1] [INFO] [1773452439.988621389] [group6.Hiro]: Hello Hiro!
minimal_param_node-1] [INFO] [1773452440.988739680] [group6.Hiro]: Hello Hiro!
minimal_param_node-1] [INFO] [1773452441.988317216] [group6.Hiro]: Hello Hiro!
minimal_param_node-3] [INFO] [1773452442.479176520] [group0.NA]: Hello Ian!
minimal_param_node-1] [INFO] [1773452442.988692463] [group6.Hiro]: Hello Hiro!
minimal_param_node-1] [INFO] [1773452443.988576149] [group6.Hiro]: Hello Hiro!
minimal_param_node-3] [INFO] [1773452444.979211901] [group0.NA]: Hello Ian!
minimal_param_node-1] [INFO] [1773452444.988707819] [group6.Hiro]: Hello Hiro!
minimal_param_node-1] [INFO] [1773452445.988289155] [group6.Hiro]: Hello Hiro!
minimal_param_node-1] [INFO] [1773452446.988291382] [group6.Hiro]: Hello Hiro!
minimal_param_node-3] [INFO] [1773452447.479541701] [group0.NA]: Hello Ian!
minimal_param_node-1] [INFO] [1773452447.989151038] [group6.Hiro]: Hello Hiro!

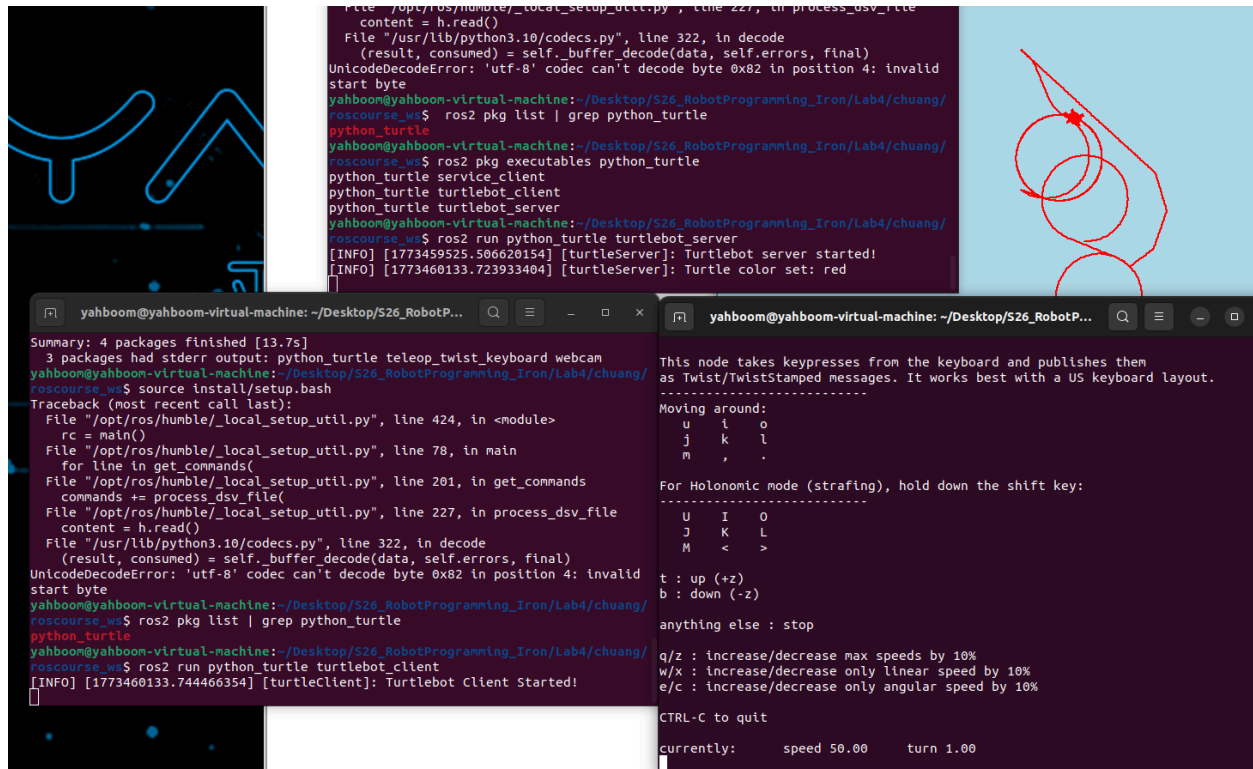
```

Task 1 - Step 17

Required Question: What do you observe when trying to move the turtle? What are possible reasons why?

- The turtle could only either rotate or move forward/backwards at a time, but not smoothly do both at the same time.

- Bc teleop_twist_keyboard is designed for real robots with small velocities, python turtle, increasing speed defaults (translation and turning) may fix this because turning/translation might be noticeable but translation is tiny, so may not be seen.
- Or teleop node may be publishing one component at a time.
- Movement felt too slow originally at 0.5 speed, or too fast 100+ speed.



```

File "/opt/ros/humble/_local_setup_util.py", line 227, in process_dsv_file
    content = h.read()
File "/usr/lib/python3.10/codecs.py", line 322, in decode
    (result, consumed) = self._buffer_decode(data, self.errors, final)
UnicodeDecodeError: 'utf-8' codec can't decode byte 0x82 in position 4: invalid
start byte
yahboom@yahboom-virtual-machine:~/Desktop/S26_RobotProgramming_Iron/Lab4/chuang/
roscourse_ws$ ros2 pkg list | grep python_turtle
python_turtle
yahboom@yahboom-virtual-machine:~/Desktop/S26_RobotProgramming_Iron/Lab4/chuang/
roscourse_ws$ ros2 pkg executables python_turtle
python_turtle service_client
python_turtle turtlebot_client
python_turtle turtlebot_server
yahboom@yahboom-virtual-machine:~/Desktop/S26_RobotProgramming_Iron/Lab4/chuang/
roscourse_ws$ ros2 run python_turtle turtlebot_server
[INFO] [1773459525.506620154] [turtleServer]: Turtlebot server started!
[INFO] [1773460133.723933404] [turtleServer]: Turtle color set: red

Summary: 4 packages finished [13.7s]
3 packages had stderr output: python_turtle teleop_twist_keyboard webcam
yahboom@yahboom-virtual-machine:~/Desktop/S26_RobotProgramming_Iron/Lab4/chuang/
roscourse_ws$ source install/setup.bash
Traceback (most recent call last):
  File "/opt/ros/humble/_local_setup_util.py", line 424, in <module>
    rc = main()
  File "/opt/ros/humble/_local_setup_util.py", line 78, in main
    for line in get_commands():
  File "/opt/ros/humble/_local_setup_util.py", line 201, in get_commands
    commands += process_dsv_file()
  File "/opt/ros/humble/_local_setup_util.py", line 227, in process_dsv_file
    content = h.read()
  File "/usr/lib/python3.10/codecs.py", line 322, in decode
    (result, consumed) = self._buffer_decode(data, self.errors, final)
UnicodeDecodeError: 'utf-8' codec can't decode byte 0x82 in position 4: invalid
start byte
yahboom@yahboom-virtual-machine:~/Desktop/S26_RobotProgramming_Iron/Lab4/chuang/
roscourse_ws$ ros2 pkg list | grep python_turtle
python_turtle
yahboom@yahboom-virtual-machine:~/Desktop/S26_RobotProgramming_Iron/Lab4/chuang/
roscourse_ws$ ros2 run python_turtle turtlebot_client
[INFO] [1773460133.744466354] [turtleClient]: Turtlebot Client Started!

This node takes keypresses from the keyboard and publishes them
as Twist/TwistStamped messages. It works best with a US keyboard layout.
-----
Moving around:
u l o
j k l
m , .

For Holonomic mode (strafing), hold down the shift key:
-----
U I O
J K L
M < >

t : up (+z)
b : down (-z)

anything else : stop

q/z : increase/decrease max speeds by 10%
w/x : increase/decrease only linear speed by 10%
e/c : increase/decrease only angular speed by 10%

CTRL-C to quit

currently:      speed 50.00      turn 1.00

```

Task 2 - Step 13

turtleColor:=green

```
yahboom@yahboom-virtual-machine: ~/Desktop/S26_RobotP...
ng...
^CTraceback (most recent call last):
  File "/home/yahboom/Desktop/S26_RobotProgramming_Iron/Lab4/chuang/roscourse_ws
/install/python_turtle/lib/python3.10/site-packages/python_turtle/turtlebot_client", line 33, in <module>
    sys.exit(load_entry_point('python-turtle==0.0.0', 'console_scripts', 'turtle
bot_client'))()
  File "/home/yahboom/Desktop/S26_RobotProgramming_Iron/Lab4/chuang/roscourse_ws
/install/python_turtle/lib/python3.10/site-packages/python_turtle/turtlebot_client.py", line 127, in main
    cli_obj = TurtleClient()
  File "/home/yahboom/Desktop/S26_RobotProgramming_Iron/Lab4/chuang/roscourse_ws
/install/python_turtle/lib/python3.10/site-packages/python_turtle/turtlebot_client.py", line 57, in __init__
    while not self.color_cli.wait_for_service(timeout_sec=1.0):
  File "/opt/ros/humble/local/lib/python3.10/dist-packages/rclpy/client.py", line 177, in wait_for_service
    time.sleep(sleep_time)
KeyboardInterrupt
[ros2run]: Interrupt
yahboom@yahboom-virtual-machine: ~/Desktop/S26_RobotProgramming_Iron/Lab4/chuang/
roscourse_ws$ ros2 run python_turtle turtlebot_client --ros-args --param turtlec
olor:=green
[INFO] [1773505739.160451492] [turtleclient]: Turtlebot Client Started!

yahboom@yahboom-virtual-machine: ~/Desktop/S26_RobotP...
start byte
yahboom@yahboom-virtual-machine: ~/Desktop/S26_RobotProgramming_Iron/Lab4/chuang/
roscourse_ws$ ros2 run python_turtle turtlebot_server
Package 'python_turtle' not found
yahboom@yahboom-virtual-machine: ~/Desktop/S26_RobotProgramming_Iron/Lab4/chuang/
roscourse_ws$ source install/setup.bash
Traceback (most recent call last):
  File "/opt/ros/humble/_local_setup_util.py", line 424, in <module>
    rc = main()
  File "/opt/ros/humble/_local_setup_util.py", line 78, in main
    for line in get_commands():
  File "/opt/ros/humble/_local_setup_util.py", line 201, in get_commands
    commands += process_dsv_file(
  File "/opt/ros/humble/_local_setup_util.py", line 227, in process_dsv_file
    content = h.read()
  File "/usr/lib/python3.10/codecs.py", line 322, in decode
    (result, consumed) = self._buffer_decode(data, self.errors, final)
UnicodeDecodeError: 'utf-8' codec can't decode byte 0x82 in position 4: invalid
start byte
yahboom@yahboom-virtual-machine: ~/Desktop/S26_RobotProgramming_Iron/Lab4/chuang/
roscourse_ws$ ros2 run python_turtle turtlebot_server
[INFO] [1773505732.034435704] [turtleServer]: Turtlebot server started!
[INFO] [1773505739.136121047] [turtleServer]: Turtle color set: green

Moving around:
u i o
j k l
m , .

For Holonomic mode (strafting), hold down the shift key:
-----
U I O
J K L
M < >

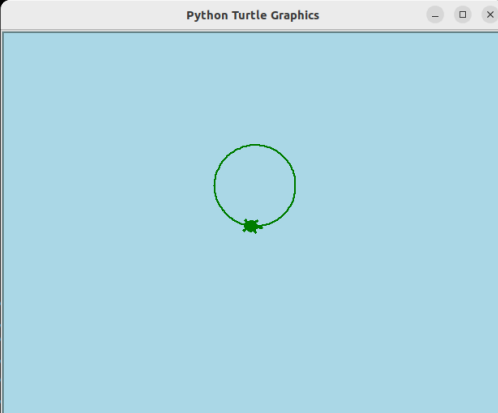
t : up (+z)
b : down (-z)

anything else : stop

q/z : increase/decrease max speeds by 10%
w/x : increase/decrease only linear speed by 10%
e/c : increase/decrease only angular speed by 10%

CTRL-C to quit

currently: speed 50.00 turn 1.00

Python Turtle Graphics

```

Task 2 - Step 14

penSize =6

```
yahboom@yahboom-virtual-machine: ~/Desktop/S26_RobotP...
/Install/python_turtle/lib/python3.10/site-packages/python_turtle/turtlebot_client.py", line 131, in main
    cli_obj.update()
  File "/home/yahboom/Desktop/S26_RobotProgramming_Iron/Lab4/chuang/roscourse_ws/Install/python_turtle/lib/python3.10/site-packages/python_turtle/turtlebot_client.py", line 87, in update
    self.turtle_display.seth(math.degrees(yaw))
  File "/usr/lib/python3.10/turtle.py", line 1936, in setheading
    self._rotate(angle)
  File "/usr/lib/python3.10/turtle.py", line 3279, in _rotate
    self.update()
  File "/usr/lib/python3.10/turtle.py", line 2663, in update
    screen._delay(screen._delayvalue) # TurtleScreenBase
  File "/usr/lib/python3.10/turtle.py", line 565, in _delay
    self.cv.after(delay)
  File "/usr/lib/python3.10/tkinter/_init_.py", line 834, in after
    self.tk.call('after', ms)
KeyboardInterrupt
[ros2run]: Interrupt
yahboom@yahboom-virtual-machine: ~/Desktop/S26_RobotProgramming_Iron/Lab4/chuang/roscourse_ws$ ros2 run python_turtle turtlebot_client --ros-args --param turtle_color:=green --param penSize:=6
[INFO] [1773506150.841031705] [turtleClient]: Turtlebot Client Started!
```

```
yahboom@yahboom-virtual-machine: ~/Desktop/S26_RobotP...
yahboom@yahboom-virtual-machine: ~/Desktop/S26_RobotProgramming_Iron/Lab4/chuang/roscourse_ws$ ros2 run python_turtle turtlebot_server
Package 'python_turtle' not found
yahboom@yahboom-virtual-machine: ~/Desktop/S26_RobotProgramming_Iron/Lab4/chuang/roscourse_ws$ source install/setup.bash
Traceback (most recent call last):
  File "/opt/ros/humble/_local_setup_util.py", line 424, in <module>
    rc = main()
  File "/opt/ros/humble/_local_setup_util.py", line 78, in main
    for line in get_commands():
  File "/opt/ros/humble/_local_setup_util.py", line 201, in get_commands
    commands += process_dsv_file(
  File "/opt/ros/humble/_local_setup_util.py", line 227, in process_dsv_file
    content = h.read()
  File "/usr/lib/python3.10/codecs.py", line 322, in decode
    (result, consumed) = self._buffer_decode(data, self.errors, final)
UnicodeDecodeError: 'utf-8' codec can't decode byte 0x82 in position 4: invalid start byte
yahboom@yahboom-virtual-machine: ~/Desktop/S26_RobotProgramming_Iron/Lab4/chuang/roscourse_ws$ ros2 run python_turtle turtlebot_server
[INFO] [1773505732.034435704] [turtleServer]: Turtlebot server started!
[INFO] [1773505739.136121047] [turtleServer]: Turtle color set: green
[INFO] [1773506150.821080611] [turtleServer]: Turtle color set: green
```

```

Moving around:
U I O
J K L
M < >

For Holonomic mode (strafing), hold down the shift key:
-----
t : up (+z)
b : down (-z)

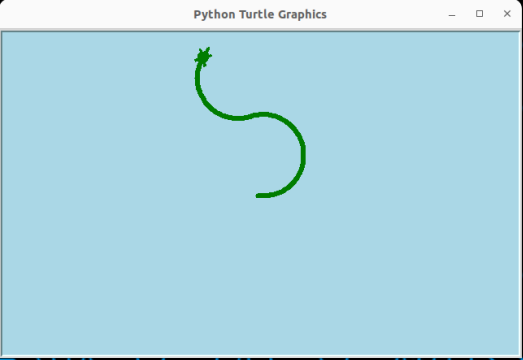
anything else : stop

q/z : increase/decrease max speeds by 10%
w/x : increase/decrease only linear speed by 10%
e/c : increase/decrease only angular speed by 10%

CTRL-C to quit

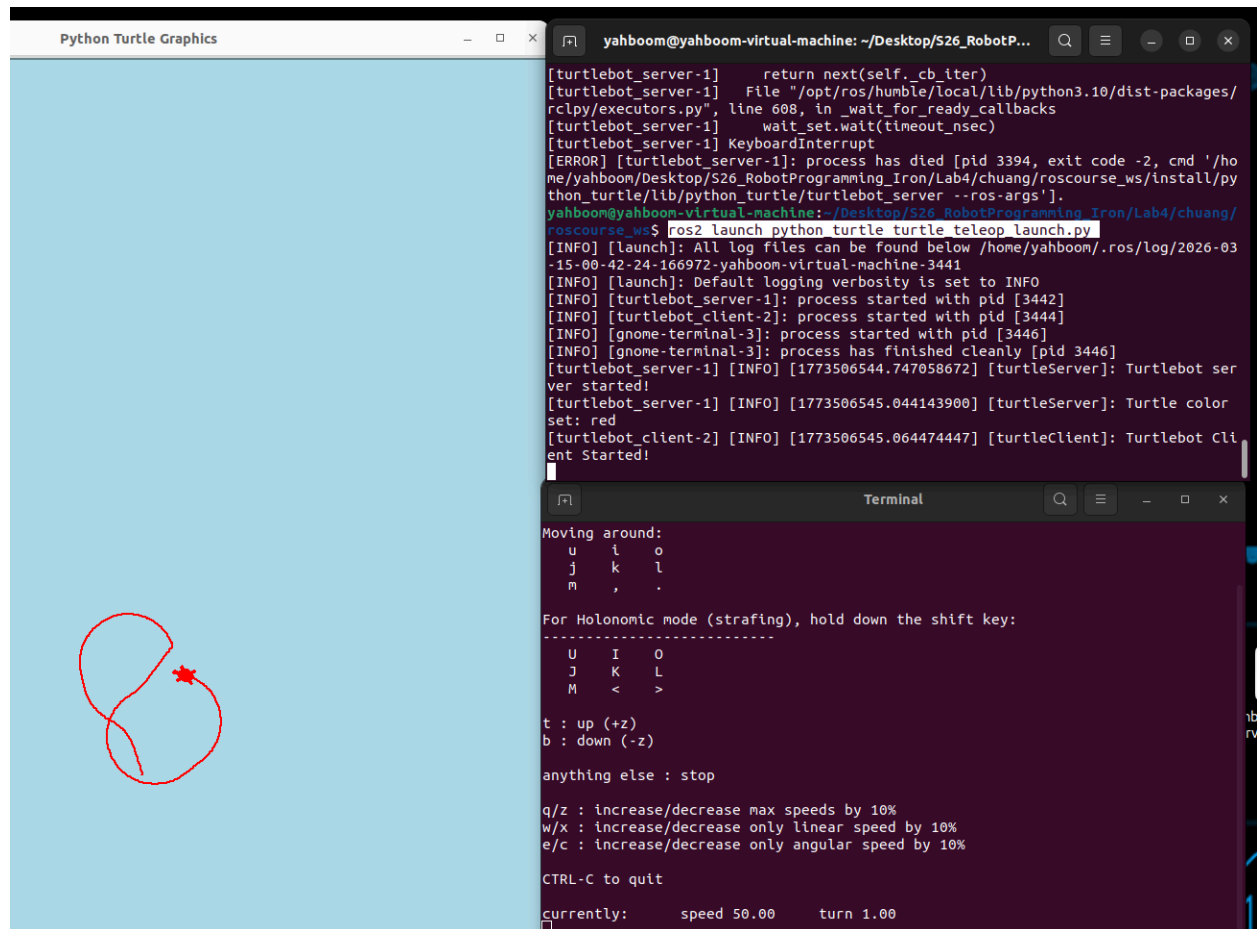
currently: speed 50.00 turn 1.00
```

```
Python Turtle Graphics
```



Task 3 - Step 8

Running everything with the command `ros2 launch python_turtle turtleteleop_launch.py`



```
[turtlebot_server-1] return next(self.cb.iter)
[turtlebot_server-1] File "/opt/ros/humble/local/lib/python3.10/dist-packages/
rclpy/executors.py", line 608, in _wait_for_ready_callbacks
[turtlebot_server-1] wait_set.wait(timeout_nsec)
[turtlebot_server-1] KeyboardInterrupt
[ERROR] [turtlebot_server-1]: process has died [pid 3394, exit code -2, cmd '/ho
me/yahboom/Desktop/S26_RobotProgramming_Iron/Lab4/chuang/roscourse_ws/install/py
thon_turtle/lib/python_turtle/turtlebot_server --ros-args'].
yahboom@yahboom-virtual-machine:~/Desktop/S26_RobotProgramming_Iron/Lab4/chuang/
roscourse_ws$ ros2 launch python_turtle turtle_teleop_launch.py
[INFO] [launch]: All log files can be found below /home/yahboom/.ros/log/2026-03
-15-00-42-24-166972-yahboom-virtual-machine-3441
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [turtlebot_server-1]: process started with pid [3442]
[INFO] [turtlebot_client-2]: process started with pid [3444]
[INFO] [gnome-terminal-3]: process started with pid [3446]
[INFO] [gnome-terminal-3]: process has finished cleanly [pid 3446]
[turtlebot_server-1] [INFO] [1773506544.747058672] [turtleServer]: Turtlebot ser
ver started!
[turtlebot_server-1] [INFO] [1773506545.044143900] [turtleServer]: Turtle color
set: red
[turtlebot_client-2] [INFO] [1773506545.064474447] [turtleClient]: Turtlebot Cli
ent Started!

Moving around:
u    i    o
j    k    l
m    ,    .

For Holonomic mode (strafing), hold down the shift key:
-----
U    I    O
J    K    L
M    <    >

t : up (+z)
b : down (-z)

anything else : stop

q/z : increase/decrease max speeds by 10%
w/x : increase/decrease only linear speed by 10%
e/c : increase/decrease only angular speed by 10%

CTRL-C to quit

currently:      speed 50.00      turn 1.00
```

Whats included in the launch file?

- A list of ROS2 node() actions that define which executables to start
- Package name and executable name for each node
- Option node config (params and output behavior)

Task 4

```

yahboom@yahboom-virtual-machine: ~/Desktop/S26_RobotP...
turtlesim.srv.SetPen_Response()

yahboom@yahboom-virtual-machine: ~/Desktop/S26_RobotProgramming_Iron/Lab4/chuang/
roscourse_ws/src/python_turtle$ ros2 service call /turtle2/set_pen turtlesim/srv
/SetPen "{r: 255, g: 0, b: 0, width: 3, 'off': 0}"
waiting for service to become available...
requester: making request: turtlesim.srv.SetPen_Request(r=255, g=0, b=0, width=3
, off=0)

response:
turtlesim.srv.SetPen_Response()

yahboom@yahboom-virtual-machine: ~/Desktop/S26_RobotProgramming_Iron/Lab4/chuang/
roscourse_ws/src/python_turtle$ ros2 service call /turtle2/set_pen turtlesim/srv
/SetPen "{r: 0, g: 255, b: 0, width: 3, 'off': 0}"
waiting for service to become available...
requester: making request: turtlesim.srv.SetPen_Request(r=0, g=255, b=0, width=3
, off=0)

response:
turtlesim.srv.SetPen_Response()

yahboom@yahboom-virtual-machine: ~/Desktop/S26_RobotProgramming_Iron/Lab4/chuang/
roscourse_ws/src/python_turtle$

[INFO] [1773507744.349379009] [turtlesim]: Spawning turtle [turtle2] at x=[0.000
000], y=[0.000000], theta=[0.000000]
[WARN] [1773507932.624377928] [turtlesim]: Rotation goal received before a previ
ous goal finished. Aborting previous goal
[WARN] [1773507932.641050042] [turtlesim]: Rotation goal received before a previ
ous goal finished. Aborting previous goal
[WARN] [1773507932.656541977] [turtlesim]: Rotation goal received before a previ
ous goal finished. Aborting previous goal
[WARN] [1773507932.673107786] [turtlesim]: Rotation goal received before a previ
ous goal finished. Aborting previous goal
[WARN] [1773507932.688421880] [turtlesim]: Rotation goal received before a previ
ous goal finished. Aborting previous goal
[WARN] [1773507932.704816172] [turtlesim]: Rotation goal received before a previ
ous goal finished. Aborting previous goal
[WARN] [1773507932.720523737] [turtlesim]: Rotation goal received before a previ
ous goal finished. Aborting previous goal
[WARN] [1773507932.736843103] [turtlesim]: Rotation goal received before a previ
ous goal finished. Aborting previous goal
[WARN] [1773507932.753179766] [turtlesim]: Rotation goal received before a previ
ous goal finished. Aborting previous goal
[WARN] [1773507932.768564849] [turtlesim]: Rotation goal received before a previ
ous goal finished. Aborting previous goal
[INFO] [1773507934.801075471] [turtlesim]: Rotation goal completed successfully

yahboom@yahboom-virtual-machine: ~/Desktop/S26_RobotP...
File "/opt/ros/h...
line 608, in _wait...
wait_set.wait(
KeyboardInterrupt
[ros2run]: Interru
yahboom@yahboom-vi
roscourse_ws$ ros2
waiting for servic
requester: making
name='turtle2')

response:
turtlesim.srv.Spaw

yahboom@yahboom-vi
roscourse_ws$ ros2
md_vel:=/turtle2/c
Reading from keybo
-----
Use arrow keys to
Use G[B|V|C|D|E|R|
ation.
'Q' to quit.

```

Why is the ability to remap so useful? Because remapping allows us to switch quickly between which turtle is being controlled, while using the same controls for each turtle.

- For task 1: Provide the screenshot of the terminals from [step 17](#)
- For task 2: Provide the modified turtlebot_client.py file with appropriate comments and formatting from [step 14](#).
- For task 2: Show the turtle from [step 13](#). It should be a different color than the default parameter value.
- For task 3: Provide a screenshot of the terminals from [step 8](#).
- For task 4: Provide a screenshot of the window with multiple turtles that have moved with different colors as well as a screenshot (one combined is fine) showing the terminals performing remote control.